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PHYSICS 4300

12/18/2024

Final Project: Timer

Abstract

The purpose of this project is to design a circuit that allows users to input a time using a keyboard and activate a timer that counts up to the specified value. The allowed input ranges from 1 second to 60 minutes. The system comprises two circuits: one for the keyboard interface and another for the timer, connected through logic components. Key components used in the circuit include 74F148N encoders, 7448N decoders, 7447N decoders, and 7490N decade counters. These components are integrated with XNOR, AND, OR, and NOT logic gates. The circuit outputs are displayed on seven-segment displays.

Methods

Keyboard Circuit

To create the keyboard circuit, we started by placing four subcircuits, each designed to handle a part of the keyboard functionality. Four subcircuits were necessary because we needed two seven-segment displays for seconds and two for minutes. We placed four output connectors to link the subcircuits to decoders and ten input connectors to connect to the DSWPK_9 switch. We began by connecting the DSWPK_9 to the nine input connectors of the subcircuits. The inputs connected to a 5V VCC power supply with 10K resistors between the input connectors and the power source. The DSWPK_9 was grounded outside of the subcircuit. Next, we connected pins 0–7 of the first 74N148N encoder chip to the switches and resistors, grounding pin EI. Pins A1 and A2 on the encoder connected to NOT gates, which then linked to output connectors, allowing the decoder to operate outside the subcircuit. Pin A0 was first connected to a NOT gate and then to the bottom pin of an OR gate, labeled U21, which was then linked to pin A on the decoder. The top pin of U21 connected back to an input connector through a NOT gate. U21 then connected to an output connector leading to pin A on the decoder. We added another 74N148N encoder and connected pins 0–3 and 7 to the VCC 5V power supply. Pin 6 was connected to an input connector. Pin A0 was then linked to the bottom pin of an OR gate labeled U23. The top pin of U23 connected to the same top wire that connects to U21 after the NOT gate. U23 connected to pin D on the decoder through an output connector. The bottom three pins on the decoder were connected to the power supply. Finally, the output pins on the 7448N decoder were connected to the SEVEN_SEG_DECIMAL_COM_K display. This process was then repeated for all four seven-segment displays, with the first two representing minutes and the last two representing seconds. A diagram of one subcircuit is shown in Figure 1, while Figure 2 illustrates its output.

Figure 1

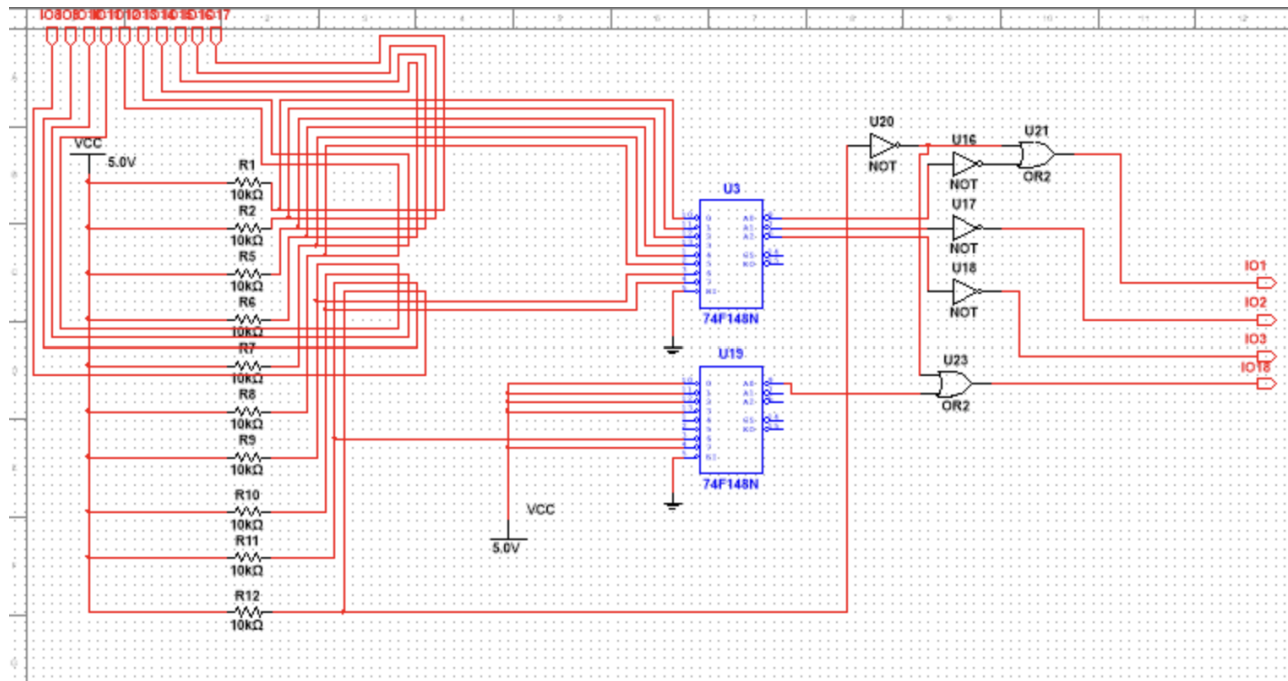
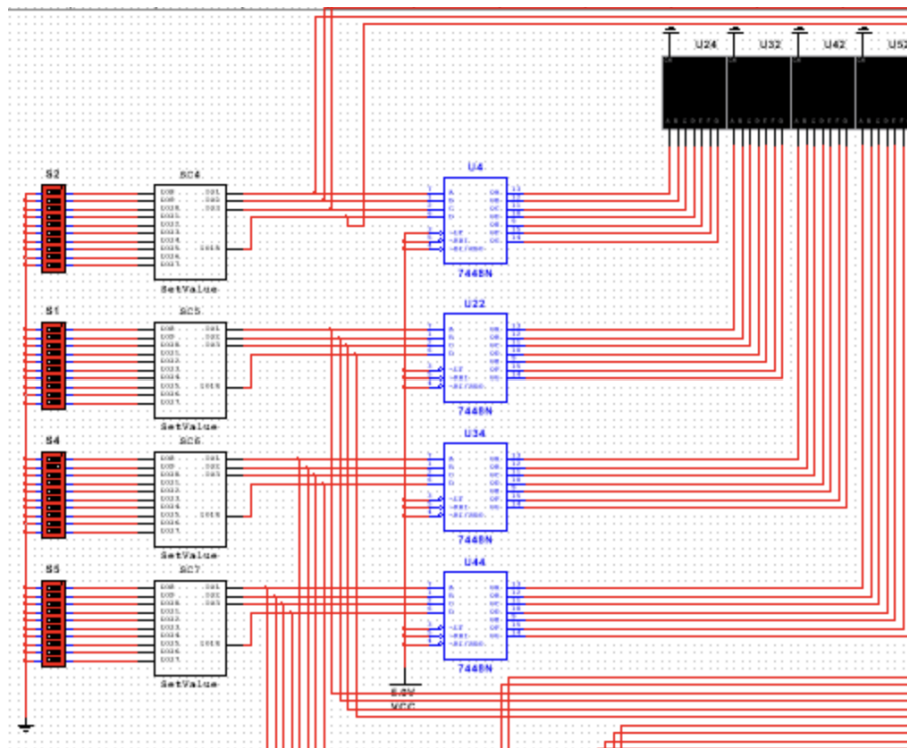


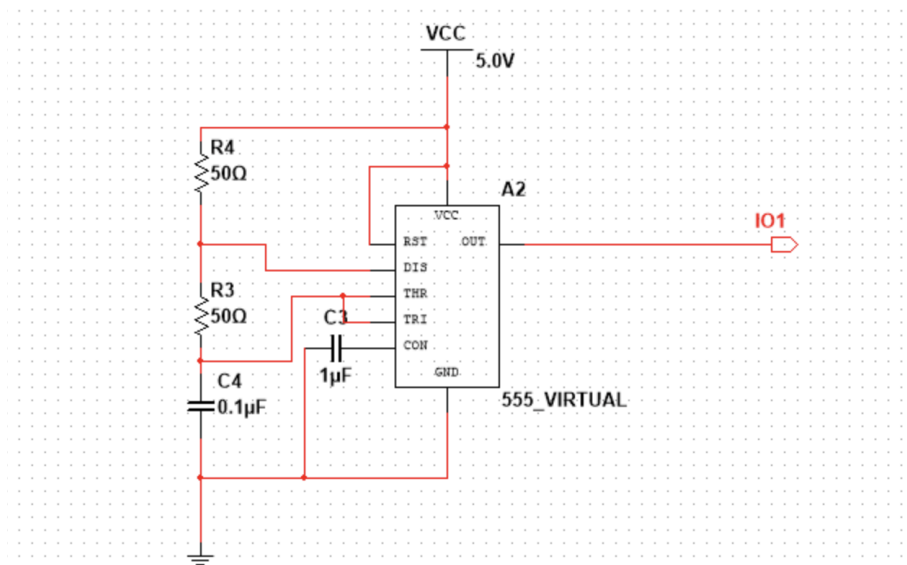
Figure 2



Timer Circuit

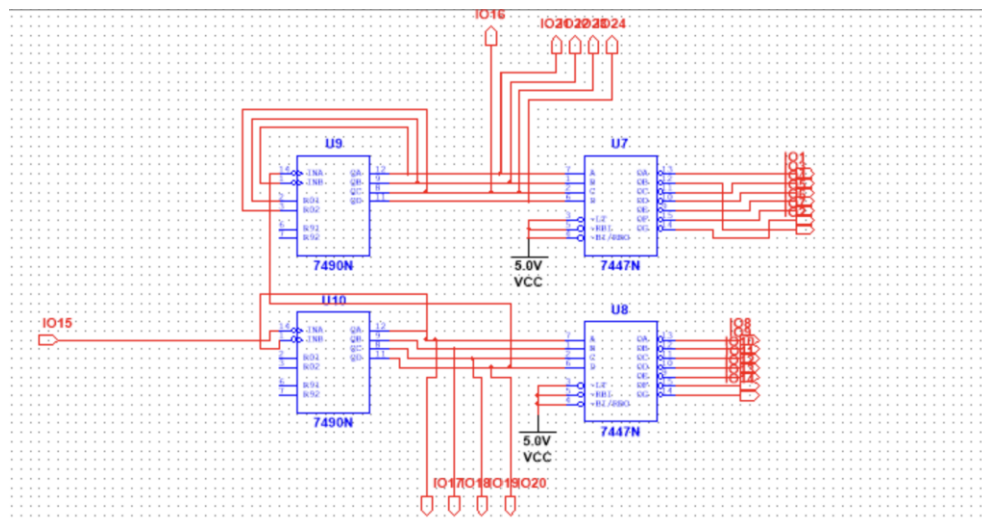
To create the timer circuit, we placed two subcircuits: one to count seconds and another to count minutes. Each subcircuit controlled two seven-segment displays. We started by connecting a 555-timer circuit to the INA input on a 7490N decade counter in the seconds subcircuit. The configuration of the 555-timer circuit is shown in Figure 3.

Figure 3



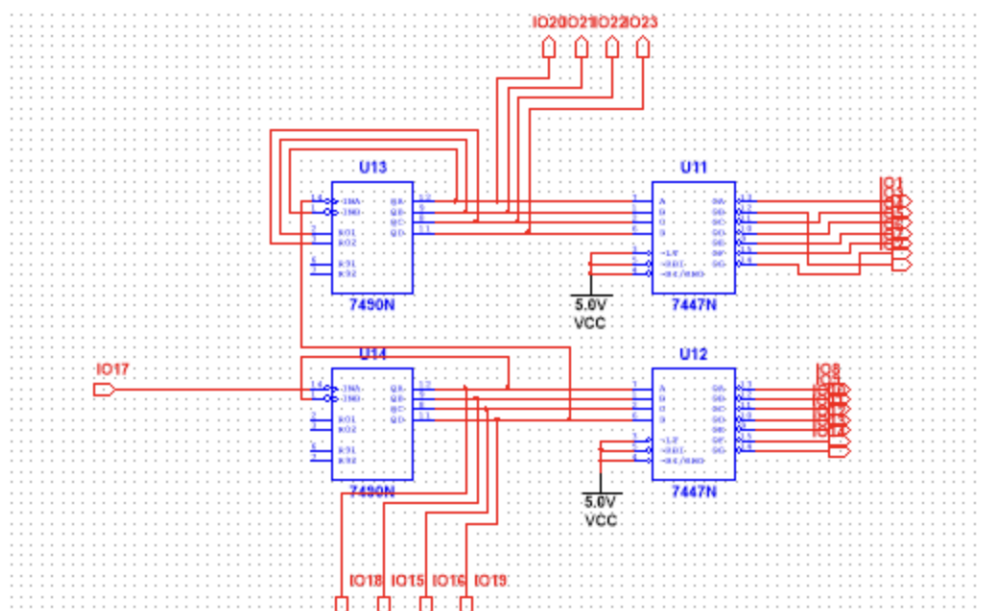
The 555-timer was placed in its own subcircuit, serving as the clock to drive the counter. We then connected the three outputs of the decade counter to a 7447N decoder, which controlled a seven-segment display. INB was connected to the first output pin of the decade counter. A second 7490N decade counter was added, with its INA input connected to the last output pin of the first counter. This setup allowed the second counter to increase by one each time the first counter completed a cycle from 0 to 9. The second counter was also connected to a 7447N decoder, which drove a second seven-segment display. The configuration of the seconds subcircuit is depicted in Figure 4.

Figure 4



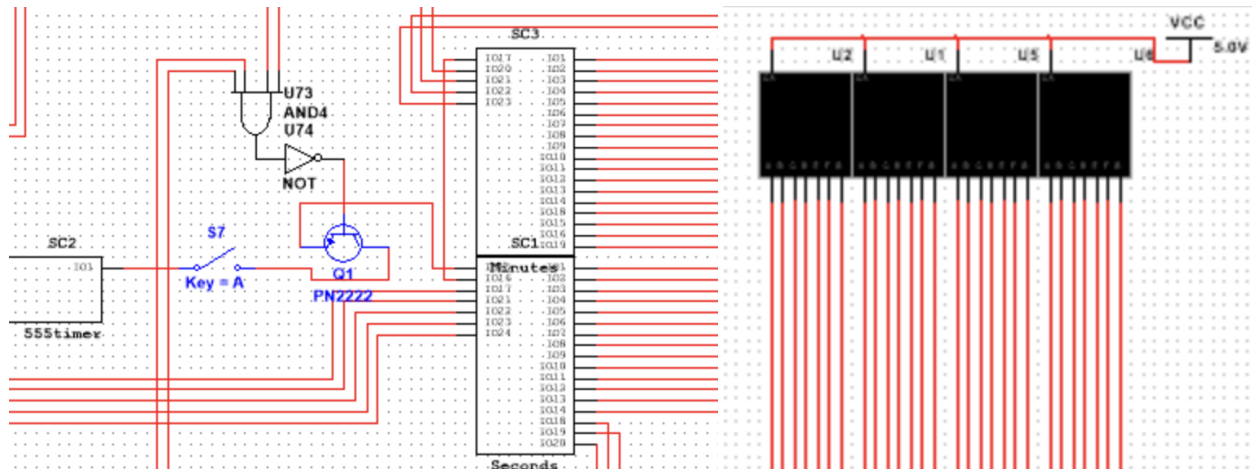
This process was repeated for the minutes subcircuit. The two subcircuits were connected by linking the INA input of the first 7490N decade counter in the minutes subcircuit to the third output pin of the second 7490N counter in the seconds subcircuit. This ensured the minutes counter incremented by one every time the seconds counter reached 60. Figure 5 shows the minutes subcircuit.

Figure 5



As you can see the two circuits are the same with the exception of IO17 in Figure 5 connecting to IO16 on Figure 4. The entirety of the counter circuit can be seen in Figure 6.

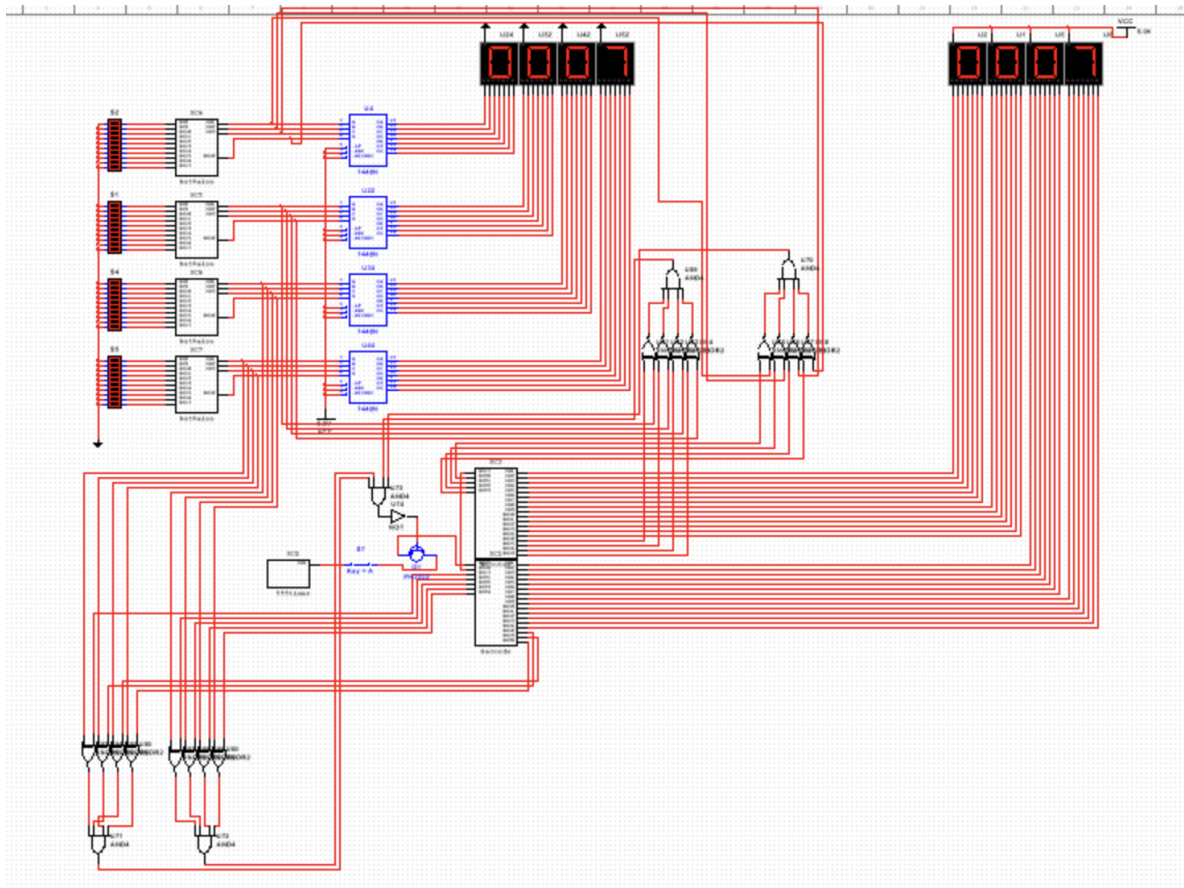
Figure 6



Combining the Two

To combine the two circuits so that the counter matched the input values from the keyboard, we decided to compare the binary outputs of each of the four displays. This was achieved by linking each display sequentially: the first seven-segment display on the keyboard circuit to the first seven-segment display on the counter circuit, and so on, using XNOR gates. We specifically connected the XNOR gates to the outputs of the 74F148N encoder and the 7490N decade counter, ensuring that there were four XNOR gates per seven-segment display. The outputs of the XNOR gates were then fed into an AND gate, which in turn was connected to a NOT gate. The NOT gate controlled a transistor placed between the 555-timer subcircuit and the seconds counter subcircuit. This configuration effectively blocked the signal from the 555-timer when the outputs of all four seven-segment displays matched. The complete circuit is shown in Figure 7.

Figure 7



Discussion

Discussing the keypad circuit, how the first encoder is connected is straightforward. Each input (0–7) connects to a segment on the DSWPK_9 switch. The encoder connects to the A, B, and C inputs on the decoder, and the decoder connects to the display. If displaying 0–7 is the desired output, the circuit is complete, requiring only 3 bits to display those numbers. However, to display 8 and 9, 4 bits are needed, necessitating an additional encoder.

We consulted the truth table of the encoder to determine the correct logic for making the two encoders work together. With 8 represented as 1000 in binary and 9 as 1001 in binary, we needed the first encoder to output L, H, H and the second encoder to output H when the switch for 8 is flipped and L when the switch for 9 is flipped. These outputs would ordinarily be opposite, but the use of NOT gates flips the H and L values appropriately.

The counter circuit is simplified by the decade counter chip, which iterates from 0 to 9 before resetting. The frequency of iteration is regulated by the 555 timer, which outputs H and L signals at consistent intervals. These binary outputs are decoded using the 7447N decoder chip. This chip, composed of flip-flops and logic gates, resets the count at 9 or whenever RO1 and RO2 receive an H signal. In our case, the chip resets at six, with those inputs connected to the two middle outputs of the chip.

The chip iterates with INA and INB inputs, which must alternate between H and L. This requirement is met by connecting INB to the first output of the chip.

To stop the counter at the set time, we compared the binary outputs of the keyboard circuit and the counter circuit. While we were unsure if this was the most efficient method, it was the most direct. XNOR gates output a 1 when both inputs are either 0 or 1. By combining these outputs with an AND gate, we ensured that the binary outputs of each chip matched.

A transistor was placed between the 555-timer and the counter circuits, with the AND gate connected to the third input of the transistor. This setup allowed us to disconnect the 555-timer, effectively shutting off the counter when the set time was reached. This process required careful wiring, particularly in the minutes and seconds counter subcircuits, where output connectors were needed to route both 7490N and 74F148N outputs to the logic gate. Overall, however, connecting the two circuits was a manageable task.

Conclusion

We successfully created a circuit that allows the user to input up to four values representing minutes and seconds and actuate a switch to start a timer that counts up to the set time. This circuit is versatile and can be applied to devices like ovens, alarm clocks, or any application requiring precise timing. Overall, the project was relatively simple, aided by the discovery of the decade counter chips, which allowed us to add more features to the circuit despite the project's time constraints.